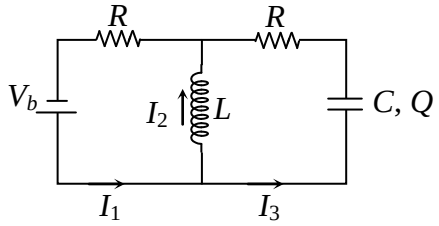
PHYSICS

SECTION – I
Single Correct Choice Type

This section contains 4 single correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

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1. In the given the circuit, which of the following choices is a possible system of equations representing the complete Kirchoff's law description of the circuit?



(a) $\begin{cases} V_b = I_1 R \\ Q = CV_b \\ V_b = -L\dfrac{dI_2}{dt} \end{cases}$

(b) $\begin{cases} V_b - L\dfrac{dI_2}{dt} - RI_1 = 0 \\ -\dfrac{Q}{C} - R(I_1 - I_2) + L\dfrac{dI_2}{dt} = 0 \\ I_3 = I_1 - I_2 \end{cases}$

(c) $\begin{cases} V_b - L\dfrac{dI_2}{dt} - RI_1 = 0 \\ -QC - R(I_1 + I_2) + L\dfrac{dI_2}{dt} = 0 \end{cases}$

(d) $\begin{cases} V_b - L\dfrac{dI_2}{dt} + RI_1 = 0 \\ -\dfrac{Q}{C} - RI_3 - L\dfrac{dI_2}{dt} = 0 \\ I_1 = I_2 + I_3 \end{cases}$

2. A steady non-turbulent stream of water comes out of a tap and falls vertically downward. As it does so, the diameter of the stream appears to get smaller. The primary reason will be

(a) The water's surface tension constricts the stream

(b) Air pressure, which decreases with altitude, squeezes the stream

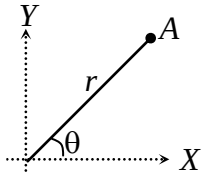
(c) The water is accelerating under gravity and so the stream must get thinner as the flow rate (velocity multiplied by cross-sectional area) must be constant.

(d) The flow does not constrict; it is an optical illusion.

3. Relative density of a material is found by ratio of weight in air and decrease in weight in water. If weight of a body in air and water are measured as $w_a = (10.0 \pm 0.1)\,\text{g}$ and $w_w = (5.0 \pm 0.1)\,\text{g}$, then percentage error in relative density will be

(a) 5 (b) 10 (c) 2.5 (d) 7.5

4. A short dipole is oriented along x-axis such that direction of dipole moment is along positive x-axis. It is observed that the magnitude of electric field and electric potential are equal at a point $\sqrt{5}\,\text{m}$ from the centre of the dipole as shown. The value of θ is



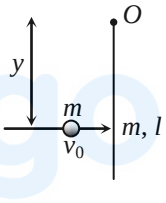
(a) 30^0 (b) 37^0 (c) 53^0 (d) 45^0

SECTION – II
Multiple Correct Choice Type

This section contains 4 multiple correct choice type questions. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONE OR MORE** is/ are correct.

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5. A uniform rod of mass m and length l is free to rotate about a fixed horizontal axis O. A small ball of mass m moving horizontally with velocity v_0 strikes the rod at depth y from O as shown in figure. If angular velocity of rod and velocity of ball just after the collision, be ω and v respectively, then the wrong statements will be



(a) If $y = l/2$, then in case of elastic collision, $v = \frac{v_0}{2}$ and $\omega = \frac{v_0}{l}$

(b) If $y = \frac{l}{2}$, then in case of elastic collision, momentum of ball-rod system remains conserved.

(c) If momentum of ball-rod system remains conserved, then $y > \frac{l}{2}$.

(d) Ball cannot be rebounded back whatever is the value of y.

6. In an African wildlife reserve a jeep is traveling at a constant speed of 105 km/hr. Alongside a cheetah is running at constant speed. The animal, frightened by the Jeep's noise, keeps a constant distance of 60m from the vehicle. Not having paid much attention to the surrounding landscape, the driver suddenly realize that the jeep is passing the very same spot where it was 60 seconds before. Then

(a) speed of cheetah is 105.4 km/hr

(b) speed of cheetah is 82.4 km/hr

(c) distance travelled by cheetah is 1373.2 m

(d) distance travelled by cheetah is 1750m

7. An ideal inductor of inductance L = 10H carrying current 5A is connected to a resistor of resistance 20Ω and mass 2 kg and specific heat 0.5 J $kg^{-1}K^{-1}$. Assume that all energy given to resistor is used to increase its temperature. Then

(a) The total change in temperature of resistor will be 125°C.

(b) The total change in temperature of resistor will be 250°C.

(c) The total charge passed through the resistor is 2.5Columb.

(d) The total charge passed through the resistor is infinite.

8. An object is taken from a point P to another point Q in a gravitational field. Then

(a) assuming the earth to be spherical, if both P and Q lie on earth's surface the work done is zero

(b) if P is on earth's surface and Q above it, the work done is minimum when it is taken along the straight line PQ

(c) the work done depends only on the position of P and Q and is independent of the path along which the particle is taken

(d) there is no net work done if the object is taken from P to Q and then brought back to P, along any path

SECTION – III
Reasoning Type

This section contains 4 reasoning type questions. Each question contains Statement-1 and Statement-2. Each question has 4 choices (a), (b), (c) and (d), out of which **ONLY ONE** is correct.

Directions: Read the following questions and choose

(A) **Statement-1** is True, **Statement-2** is True; **Statement-2** is a correct explanation for **Statement-1**.

(B) **Statement-1** is True, **Statement-2** is True; **Statement-2** is not a correct explanation for **Statement-1**.

(C) **Statement-1** is True, **Statement-2** is False.

(D) **Statement-1** is False, **Statement-2** is True.

9. **Statement-1:** In a conductor of non-uniform cross-section, the current is same, though the drift velocity is different.

Statement-2: In series current is same and for same current drift velocity is inversely proportional to cross-sectional area $\left(v_d \propto \frac{1}{A}\right)$.

(a) A (b) B (c) C (d) D

10. **Statement-1:** Two conductors are identical in shape and size but are of different material. If both conductors are placed in an identical electric field, charges induced on them may be different.

Statement-2: Induced charge $q' = -q\left(1 - \frac{1}{K}\right)$; more the value of K, more will be the induced charge.

(a) A (b) B (c) C (d) D

11. **Statement-1:** If a particle moves in xy plane it has only z-component of angular momentum.

Statement-2: Angular momentum $\vec{L} = \vec{r} \times \vec{p}$ is always perpendicular to position vector $\vec{r} = (\hat{i}x + \hat{j}y)$.

(a) A (b) B (c) C (d) D

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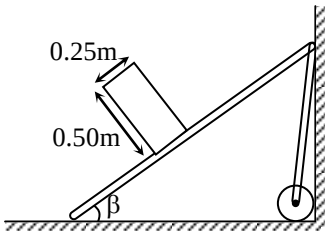
12. **Statement-1:** A race car traveling around circular track have a non zero impulse.
Statement-2: The impulse is zero only when there is no net change in momentum.
(a) (A) (b) (B) (c) (C) (d) (D)

SECTION–IV
Comprehension Type

This section contains 2 groups of questions. Each group has 3 multiple choice questions based on a paragraph. Each question has 4 choices: (a), (b), (c) and (d) for its answer, out of which **ONLY ONE** is correct.

Passage-I

An engineer is designing a conveyor system for loading lay bales into a wagon. Each bale is 0.25m wide, 0.50 m high, and 0.80 m long (the dimension perpendicular to the plane of the figure), with mass 30.0 kg. The centre of gravity of each bale is at its geometrical centre. The coefficient of static friction between a bale and the conveyor belt is 0.60, and the belt moves with constant speed. The angle β of the conveyor is slowly increased. At some critical angle (β_1) a bale will tip (if it doesn't slip first), and at some different critical angle (β_2) it will slip (if it doesn't tip first)



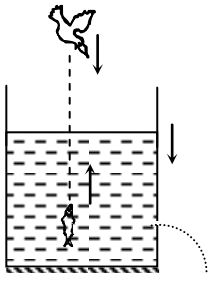
13. The first critical angle, β_1 (in the same conditions) at which it tips without slipping is
(a) $\tan^{-1}0.50$ (b) $\tan^{-1}0.60$
(c) $\tan^{-1}0.40$ (d) $\tan^{-1}0.20$

14. The second critical angle, β_2 (in the same conditions) at which it slips without tipping is
(a) $\tan^{-1}0.50$ (b) $\tan^{-1}0.60$
(c) $\tan^{-1}0.12$ (d) $\tan^{-1}0.70$

15. The first critical angle, β_1 at which it tips if the coefficient of friction were 0.40 is
(a) $\tan^{-1}0.50$ (b) $\tan^{-1}0.60$
(c) $\tan^{-1}0.70$ (d) $\tan^{-1}0.20$

Passage-II

A bird in air is diving vertically over a tank with speed 5 cm/s. Base of tank is silvered. A fish in the tank is rising upward along the same line with speed 2 cm/s. Water level is falling at rate of 2 cm/s. [Take $\mu_{water} = 4/3$]



16. Speed of the image of fish as seen by the bird directly in cm/s is
(a) 8 (b) 6 (c) 3 (d) 4

17. Speed of the image of fish formed after reflection in the mirror as seen by the bird in cm/s is
(a) 8 (b) 6 (c) 3 (d) 4

18. Speed of image of bird relative to the fish looking upwards in cm/s is
(a) 8 (b) 6 (c) 3 (d) 4

SECTION–V
Matrix–Match Type

This section contains 2 questions. Each question contains statements given in two columns, which have to be matched. The statements in **Column I** are labelled A, B, C and D, while the statements in **Column II** are labelled 1, 2, 3, 4 and 5. Any given statement in **Column I** can have correct matching with **ONE OR MORE** statement(s) in **Column II**. The appropriate bubbles corresponding to the answers to these questions have to be darkened as illustrated in the following example:

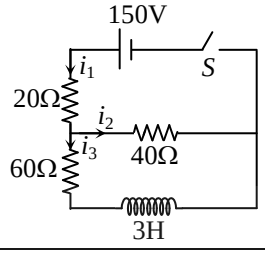
If the correct matches are A – 1, 4 and 5; B – 2 and 3; C – 1 and 2; and D – 4 and 5; then the correct darkening of bubbles will look like the following.

	1	2	3	4	5
A	●	○	○	●	●
B	○	●	●	○	○
C	●	●	○	○	○
D	○	○	○	●	●

1. The position (x) of a particle varies with time as $x = t^3 - 3t^2$, where t is time in second. Match column I with column II.

Column I	Column II
(A) Position of particle is zero at	**1.** $t = 0$ s
(B) Velocity of particle is zero at	**2.** $t = 1$s
(C) Magnitude of acceleration of particle is zero at	**3.** $t = 2$ s
(D) Magnitude of displacement is equal to distance traveled by the particle at	**4.** $t = 3$ s **5.** t = 1.5 s

2. Match the values of i_1, i_2, i_3 for the circuit shown under conditions of column I with column II.



Column -I	Column -II
(A) Soon after switch is closed	**1.** i_1 = 0 A
(B) Long after switch is closed	**2.** i_2 = 1.37 A
(C) Long after switch is reopened	**3.** i_3 = 1.37 A
(D) Soon after switch is reopened	**4.** $i_1 = i_2$ **5.** $i_2 = i_3$